

TECHNICAL UNIVERSITY OF DENMARK



02611 Optimization for Data Science

ASSIGNMENT: LOGISTIC REGRESSION

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April 24, 2026

Question 1

a) Assume that the class probabilities satisfy the two natural requirements:

$$p_1(\tau) + p_2(\tau) = 1, \quad (1)$$

$$\ln \left(\frac{p_1(\tau)}{p_2(\tau)} \right) = \tau. \quad (2)$$

From the second condition (2), exponentiating both sides yields $\frac{p_1(\tau)}{p_2(\tau)} = e^\tau$, and therefore $p_1(\tau) = e^\tau p_2(\tau)$.

Substituting this expression into the normalization condition (1) gives

$$e^\tau p_2(\tau) + p_2(\tau) = 1 \quad \Rightarrow \quad p_2(\tau)(1 + e^\tau) = 1.$$

Solving for $p_2(\tau)$ results in $p_2(\tau) = \frac{1}{1+e^\tau}$.

Consequently,

$$p_1(\tau) = 1 - p_2(\tau) = \frac{e^\tau}{1 + e^\tau}.$$

Thus, the logistic regression model

$$p_1(\tau) = \frac{e^\tau}{1 + e^\tau}, \quad p_2(\tau) = \frac{1}{1 + e^\tau}$$

is the *unique* model that satisfies both given requirements.

b) The decision boundary is defined as the set

$$\{x \in \mathbb{R}^n \mid p_1(w^T x + b) = p_2(w^T x + b)\}.$$

Let $\tau = w^T x + b$. Using the logistic regression model from (1)-(2), the equality $p_1(\tau) = p_2(\tau)$ implies

$$\frac{e^\tau}{1 + e^\tau} = \frac{1}{1 + e^\tau} \quad \Rightarrow \quad e^\tau = 1 \quad \Rightarrow \quad \tau = 0.$$

Hence,

$$w^T x + b = 0 \quad \Leftrightarrow \quad w^T x = -b.$$

This is the equation of a hyperplane in $\mathbb{R}^n$ (with normal vector $w \in \mathbb{R}^n$). Therefore, the decision boundary is a hyperplane in $\mathbb{R}^n$.

c) The logistic sigmoid function is given by $\sigma(\tau) = p_1(\tau) = \frac{e^\tau}{1+e^\tau}$. We show that $p_2(\tau) = 1 - \sigma(\tau) = \sigma(-\tau)$.

First,

$$1 - \sigma(\tau) = 1 - \frac{e^\tau}{1 + e^\tau} = \frac{1 + e^\tau - e^\tau}{1 + e^\tau} = \frac{1}{1 + e^\tau} = p_2(\tau).$$

Second,

$$\sigma(-\tau) = \frac{e^{-\tau}}{1 + e^{-\tau}} = \frac{e^{-\tau}}{1 + e^{-\tau}} \cdot \frac{e^\tau}{e^\tau} = \frac{1}{1 + e^\tau} = p_2(\tau).$$

Hence, $p_2(\tau) = 1 - \sigma(\tau) = \sigma(-\tau)$.

d) From part 1.c, we have $\sigma(-\tau) = \frac{1}{1+e^\tau}$. Hence

$$-\ln \sigma(-\tau) = -\ln \left(\frac{1}{1+e^\tau} \right) = \ln(1+e^\tau) = \phi(\tau).$$

Differentiating $\phi(\tau) = \ln(1+e^\tau)$ gives

$$\phi'(\tau) = \frac{e^\tau}{1+e^\tau} = \sigma(\tau).$$

Differentiating once more,

$$\phi''(\tau) = \frac{e^\tau}{(1+e^\tau)^2} = \frac{e^\tau}{1+e^\tau} \cdot \frac{1}{1+e^\tau} = \sigma(\tau)(1-\sigma(\tau)).$$

Since $\sigma'(\tau) = \sigma(\tau)(1-\sigma(\tau))$, we conclude that

$$\phi(\tau) = -\ln \sigma(-\tau), \quad \phi'(\tau) = \sigma(\tau), \quad \phi''(\tau) = \sigma'(\tau) = \sigma(\tau)(1-\sigma(\tau)).$$

Question 2

a) Let $\tau_i = w^T x_i + b$ and define $\tilde{y}_i = 3 - 2y_i \in \{-1, 1\}$. Then we show that $p_{y_i}(\tau_i) = \sigma(\tilde{y}_i \tau_i)$.

If $y_i = 1$, then $\tilde{y}_i = 1$, and hence $\sigma(\tilde{y}_i \tau_i) = \sigma(\tau_i) = p_1(\tau_i) = p_{y_i}(\tau_i)$.

If $y_i = 2$, then $\tilde{y}_i = -1$, and using that $\sigma(-\tau) = 1 - \sigma(\tau)$, we obtain $\sigma(\tilde{y}_i \tau_i) = \sigma(-\tau_i) = p_2(\tau_i) = p_{y_i}(\tau_i)$.

Thus, the identity holds for both $y_i \in \{1, 2\}$, and we conclude that

$$p_{y_i}(w^T x_i + b) = \sigma((3 - 2y_i)(w^T x_i + b)).$$

b) From part 2.a, with $\tilde{y}_i = 3 - 2y_i$ and $\tilde{x}_i = (x_i, 1)$, we have

$$p_{y_i}(w^T x_i + b) = \sigma(\tilde{y}_i \tilde{x}_i^T \theta),$$

where $\theta = (w, b)$. Hence,

$$\ln \mathcal{L}(\theta) = \sum_{i=1}^N \ln \sigma(\tilde{y}_i \tilde{x}_i^T \theta).$$

Using $\phi(t) = -\ln \sigma(-t)$ with $t = -\tilde{y}_i \tilde{x}_i^T \theta$, we get

$$\ln \sigma(\tilde{y}_i \tilde{x}_i^T \theta) = -\phi(-\tilde{y}_i \tilde{x}_i^T \theta).$$

Therefore,

$$\ln \mathcal{L}(\theta) = -\sum_{i=1}^N \phi(-\tilde{y}_i \tilde{x}_i^T \theta).$$

c) From part 2.b, the negative log-likelihood is

$$-\ln \mathcal{L}(\theta) = \sum_{i=1}^N \phi(-\tilde{y}_i \tilde{x}_i^T \theta).$$

From part 1.d,

$$\phi''(\tau) = \sigma(\tau)(1-\sigma(\tau)) \geq 0,$$

so ϕ is convex. Moreover, the map

$$\theta \mapsto -\tilde{y}_i \tilde{x}_i^T \theta$$

is affine in θ . Therefore, each function

$$\phi(-\tilde{y}_i \tilde{x}_i^T \theta)$$

is convex as a composition of a convex function with an affine map. Since sums of convex functions are convex, it follows that $-\ln \mathcal{L}(\theta)$ is convex in θ .

Question 3

a) From part 2.b, we have

$$-\ln \mathcal{L}(\theta) = \sum_{i=1}^N \phi(-\tilde{y}_i \tilde{x}_i^T \theta).$$

Introduce the auxiliary variable $z \in \mathbb{R}^N$ defined by $z = -\text{diag}(\tilde{y}) \tilde{X} \theta$, or componentwise $z_i = -\tilde{y}_i \tilde{x}_i^T \theta$.

Then $-\ln \mathcal{L}(\theta) = \sum_{i=1}^N \phi(z_i)$. Therefore, the original problem

$$\min_{\theta} \left\{ -\ln \mathcal{L}(\theta) + \frac{\gamma}{2} \|D\theta\|_2^2 \right\}$$

is equivalent to

$$\min_{\theta, z} \left\{ \sum_{i=1}^N \phi(z_i) + \frac{\gamma}{2} \|D\theta\|_2^2 \right\} \quad \text{s.t.} \quad z = -\text{diag}(\tilde{y}) \tilde{X} \theta,$$

The two formulations are equivalent since the constraint ensures $z_i = -\tilde{y}_i \tilde{x}_i^T \theta$ for all i .

b) The convex conjugate of $\phi(\tau) = \ln(1 + e^\tau)$ is

$$\phi^*(\rho) = \sup_{\tau} \{ \rho \tau - \phi(\tau) \}.$$

Differentiating with respect to τ and setting to zero gives

$$\rho = \frac{e^\tau}{1 + e^\tau} = \sigma(\tau),$$

which implies

$$\tau = \ln \left(\frac{\rho}{1 - \rho} \right), \quad \rho \in (0, 1).$$

Substituting into the objective yields

$$\phi^*(\rho) = \rho \ln \left(\frac{\rho}{1 - \rho} \right) + \ln(1 - \rho) = \rho \ln \rho + (1 - \rho) \ln(1 - \rho).$$

Using the convention $0 \ln 0 = 0$, this extends to $\rho \in [0, 1]$. Thus,

$$\phi^*(\rho) = -H(\rho), \quad \rho \in [0, 1],$$

and $\phi^*(\rho) = +\infty$ otherwise.

c) The constraint in (6) can be written as $z + \text{diag}(\tilde{y}) \tilde{X} \theta = 0$.

With multiplier $\lambda \in \mathbb{R}^N$, the Lagrangian is

$$\mathcal{L}(\theta, z, \lambda) = \sum_{i=1}^N \phi(z_i) + \frac{\gamma}{2} \|D\theta\|_2^2 + \lambda^T (z + \text{diag}(\tilde{y}) \tilde{X} \theta).$$

Minimizing over z gives

$$\inf_z \sum_{i=1}^N (\phi(z_i) + \lambda_i z_i) = - \sum_{i=1}^N \phi^*(-\lambda_i),$$

which is finite iff $-\lambda_i \in [0, 1]$, i.e. $\lambda_i \in [-1, 0]$.

Minimizing over θ gives the finite value

$$-\frac{1}{2\gamma} \left\| D\tilde{X}^T \text{diag}(\tilde{y})\lambda \right\|_2^2,$$

provided that $\tilde{y}^T \lambda = 0$, since the bias term is not regularized. Therefore, the dual problem is

$$\begin{aligned} \max_{\lambda \in \mathbb{R}^N} \quad & -\sum_{i=1}^N \phi^*(-\lambda_i) - \frac{1}{2\gamma} \left\| D\tilde{X}^T \text{diag}(\tilde{y})\lambda \right\|_2^2 \\ \text{s.t.} \quad & \lambda \in [-1, 0]^N, \quad \tilde{y}^T \lambda = 0. \end{aligned}$$

d) The KKT conditions for (6), with multiplier $\lambda \in \mathbb{R}^N$ for the equality constraint, are:

- Stationarity w.r.t. z : $\phi'(z_i) + \lambda_i = 0 \Rightarrow \lambda_i = -\sigma(z_i), \quad i = 1, \dots, N.$
- Stationarity w.r.t. θ : $\gamma D^T D\theta + \tilde{X}^T \text{diag}(\tilde{y})\lambda = 0.$
- Primal feasibility: $z + \text{diag}(\tilde{y})\tilde{X}\theta = 0.$
- Dual feasibility: $-1 \leq \lambda_i \leq 0, \quad i = 1, \dots, N.$

Complementary slackness does not apply, since the constraints are equalities.

e) Yes, strong duality holds. The problem in (6) is convex since ϕ is convex, the regularization term is convex, and the constraint is affine. Moreover, the problem is feasible: for any $\theta \in \mathbb{R}^{n+1}$, choosing

$$z = -\text{diag}(\tilde{y})\tilde{X}\theta$$

satisfies the equality constraint. Hence Slater's condition holds for this convex problem with affine equality constraints, and therefore strong duality holds.

f) From the KKT conditions, we have

$$\lambda_i = -\phi'(z_i) = -\sigma(z_i).$$

At optimality, $z_i^* = -\tilde{y}_i \tilde{x}_i^T \theta^*$, hence

$$\lambda_i^* = -\sigma(-\tilde{y}_i \tilde{x}_i^T \theta^*).$$

Thus,

$$-\lambda_i^* = \sigma(-\tilde{y}_i \tilde{x}_i^T \theta^*).$$

This corresponds to the probability assigned by the model to the incorrect class for sample i . Hence, the dual variables encode (up to a sign) the predicted probabilities of misclassification under the logistic model.

g) Given an optimal primal solution (θ^*, z^*) , the corresponding dual optimal point is

$$\lambda_i^* = -\phi'(z_i^*) = -\sigma(z_i^*), \quad i = 1, \dots, N.$$

Using $z_i^* = -\tilde{y}_i \tilde{x}_i^T \theta^*$, this can be written as

$$\lambda_i^* = -\sigma(-\tilde{y}_i \tilde{x}_i^T \theta^*).$$

Since the problem is convex and strong duality holds, the KKT conditions are necessary and sufficient for optimality. The above choice of λ^* satisfies the stationarity condition with respect to z , and together with primal feasibility and stationarity in θ , all KKT conditions hold. Therefore, λ^* is dual optimal.

Question 4

a) The objective function is

$$f_0(\theta) = \sum_{i=1}^N \phi(-\tilde{y}_i \tilde{x}_i^T \theta) + \frac{\gamma}{2} \|D\theta\|_2^2.$$

Let $z_i = -\tilde{y}_i \tilde{x}_i^T \theta$. Using $\phi'(z_i) = \sigma(z_i)$, the chain rule gives

$$\nabla_{\theta} \phi(z_i) = \phi'(z_i) \nabla_{\theta} z_i = \sigma(z_i) (-\tilde{y}_i \tilde{x}_i).$$

Summing over all i yields

$$\sum_{i=1}^N \sigma(z_i) (-\tilde{y}_i \tilde{x}_i) = -\tilde{X}^T \text{diag}(\tilde{y}) q(z),$$

where

$$q(z) = (\sigma(z_1), \dots, \sigma(z_N))^T.$$

Moreover, since $D^T D = D$,

$$\nabla_{\theta} \frac{\gamma}{2} \|D\theta\|_2^2 = \gamma D\theta.$$

Therefore,

$$\nabla f_0(\theta) = -\tilde{X}^T \text{diag}(\tilde{y}) q(z) + \gamma D\theta.$$

b) Differentiating the gradient from part 4.a gives

$$\nabla^2 f_0(\theta) = \tilde{X}^T \text{diag}(\sigma'(z_1), \dots, \sigma'(z_N)) \tilde{X} + \gamma D,$$

where

$$z = -\text{diag}(\tilde{y}) \tilde{X} \theta.$$

Here we used that $\text{diag}(\tilde{y})^2 = I$.

Since

$$\sigma'(\tau) = \sigma(\tau)(1 - \sigma(\tau)), \quad 0 \leq \sigma'(\tau) \leq \frac{1}{4},$$

we have

$$0 \preceq \text{diag}(\sigma'(z_1), \dots, \sigma'(z_N)) \preceq \frac{1}{4} I.$$

Therefore,

$$0 \preceq \tilde{X}^T \text{diag}(\sigma'(z_1), \dots, \sigma'(z_N)) \tilde{X} \preceq \frac{1}{4} \tilde{X}^T \tilde{X}.$$

Also, since $D \preceq I$, we have

$$\gamma D \preceq \gamma I.$$

Hence,

$$\nabla^2 f_0(\theta) \preceq \left(\frac{1}{4} \|\tilde{X}\|_2^2 + \gamma \right) I.$$

Thus,

$$\|\nabla^2 f_0(\theta)\|_2 \leq \frac{1}{4} \|\tilde{X}\|_2^2 + \gamma.$$

Therefore, ∇f_0 is Lipschitz continuous with Lipschitz constant

$$L = \gamma + \frac{1}{4} \|\tilde{X}\|_2^2.$$

Question 5

We implemented gradient descent and the accelerated gradient method with constant step size $\alpha = 1/L$, where $L = \gamma + \frac{1}{4}\|\tilde{X}\|_2^2$. The methods were applied to the breast cancer Wisconsin dataset with $\gamma = 3$.

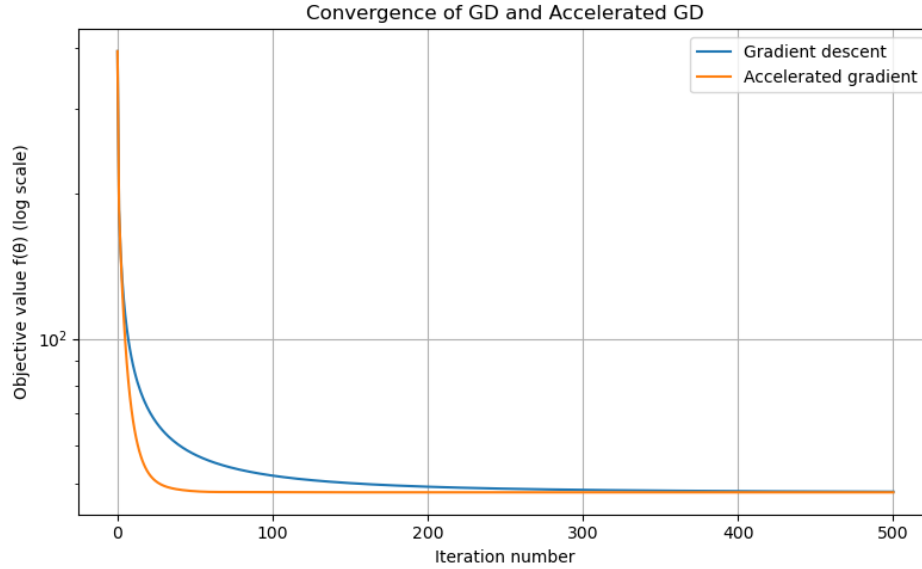


Figure 1: Convergence of $f(\theta)$ for GD and AGD (log-scale).

The convergence behavior is shown in Figure 1. Both methods decrease the objective value over iterations. However, the accelerated gradient method converges significantly faster than standard gradient descent, reaching a low objective value in far fewer iterations.

This is particularly evident in the logarithmic plot, where the accelerated method exhibits a steeper decrease in the early iterations. Both methods eventually approach the same minimum, confirming that they solve the same optimization problem.

Question 6

The trade-off curve was computed for 20 logarithmically spaced values of $\gamma \in [10^{-3}, 10^3]$ using the accelerated gradient method with warm-starting. As shown in Figure 2, increasing γ decreases $\frac{1}{2}\|D\theta^*(\gamma)\|_2^2$ while increasing the negative log-likelihood $-\ln \mathcal{L}(\theta^*(\gamma))$.

The curve exhibits a clear “L-shape”. For small γ , the model achieves better data fit (low loss) but with large weights, indicating potential overfitting. For large γ , the weights are strongly regularized, yielding a simpler model with higher loss, corresponding to underfitting.

Thus, the curve illustrates the classical bias–variance trade-off. The bend of the curve indicates a balanced region of γ , where a good compromise between model complexity and data fit is achieved. Representative values are shown in Table 1.

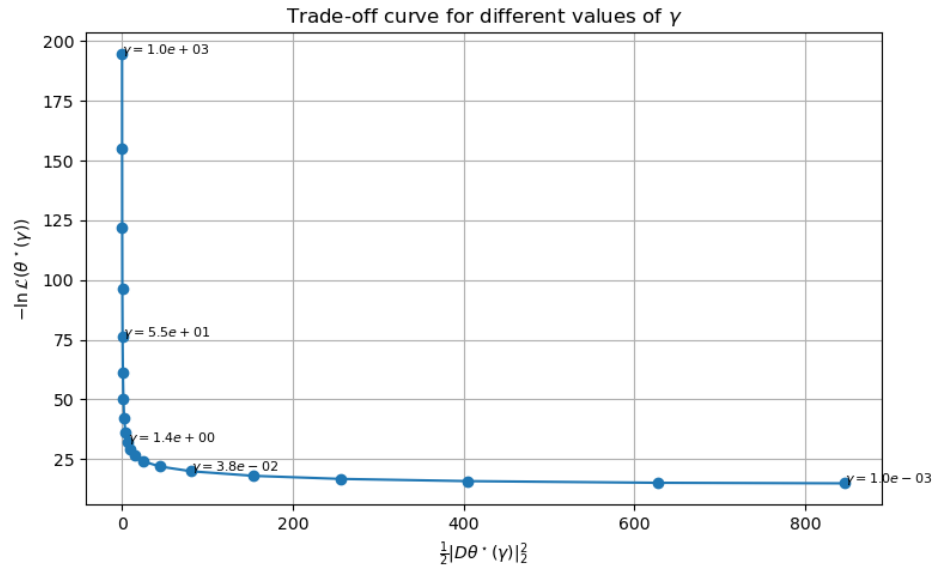


Figure 2: Trade-off curve between $\frac{1}{2}\|D\theta^*(\gamma)\|_2^2$ and $-\ln \mathcal{L}(\theta^*(\gamma))$.

γ	$\frac{1}{2}\ D\theta^*(\gamma)\ _2^2$	$-\ln \mathcal{L}(\theta^*(\gamma))$
10^{-3}	846.79	14.82
$3.79 \cdot 10^{-2}$	81.19	19.86
1.44	5.98	32.06
54.6	0.63	76.33
10^3	0.06	194.58

Table 1: Representative points on the trade-off curve.

AI Statement

AI tools were used to support understanding of theoretical concepts and to improve the clarity of the written presentation. All mathematical derivations, implementations, and results were independently developed and verified by the author.